AMANI KIRUGA

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EDUCATION

PhD in Computer Science, Harvard University — Expected 2030
Advisor: Prof. Yilun Du
Research interests: Generative models, 3D geometry, and scene understanding for robotics.

BS in Computer Science, University of Delaware — Aug. 2024
Cumulative GPA: 3.94/4.00 | Major GPA: 4.00/4.00

PUBLICATIONS

- **UGG: Unified Generative Geometry Model**
  A. Kiruga*, Z. Li*, R. Cai*, H. Lillemark, X. Wu, A. Tewari, Y. Du.
  NeurIPS 2026.
- **Infer3D: 3D Reconstruction through Inverse Generative Modelling**
  A. Kiruga, M. Hamilton, J. B. Tenenbaum, W. T. Freeman, V. Sitzmann, Y. Du, A. Tewari.
  Preprint, 2026.
- **Learning Self-Orchestration in Decentralized Agent Swarms**
  G. Xu*, Z. Qi*, **A. Kiruga**, H. Lakkaraju, Y. Du.
  Preprint, 2026.
- **A Portal for High-Precision Atomic Data and Computation**
  A. Kiruga, C. Cheung, P. Barakshan, A. Bhosale, R. Eigenmann, M. Safronova, B. Arora.
  Computer Physics Communications, 2025.
- **Infant Video Interaction Recognition using Monocular Depth Estimation**
  C. Rasmussen, **A. Kiruga**, J. Orlando, M. Lobo.
  ISVC 2024 (Oral).
- **Non-Hierarchical Transformers for Pedestrian Segmentation**
  A. Kiruga, X. Peng.
  CVPR Workshop (AVA), 2023.

*Equal contribution.

RESEARCH EXPERIENCE

PhD Researcher — Sep. 2025 – Present
Harvard University, Embodied Minds Lab — *Cambridge, MA*

- Research generative models and 3D scene understanding with Prof. Yilun Du.
- Worked on UGG, a unified generative geometry model (co-first author, NeurIPS 2026), and Infer3D, an approach to 3D reconstruction through inverse generative modelling (first-author preprint, 2026).

Student Researcher — Sep. 2023 – Mar. 2025
MIT Computer Science and Artificial Intelligence Laboratory (CSAIL) — *Cambridge, MA*

- Worked with Prof. Vincent Sitzmann and Prof. Ayush Tewari on unsupervised learning.
- Developed a method for out-of-domain generalization for single-image 3D reconstruction.

RESEARCH EXPERIENCE (CONTINUED)

Student Researcher Jan. 2024 – May 2024
University of Delaware, Dynamic Vision Lab *Newark, DE*

- Studied video action recognition and pose detection with Prof. Christopher Rasmussen. Benchmarked LLM performance and trained ResNet and DINO baselines using PyTorch.

Student Researcher Jun. 2023 – Sep. 2023
MIT Media Lab, Viral Communications Group *Cambridge, MA*

- Designed and built an interactive portal to analyze media sources with LLMs using Next.js and MongoDB, working with Dr. Andrew Lippman.

Student Researcher Jun. 2021 – May 2023
University of Delaware, D-REAL Lab *Newark, DE*

- Collected and annotated 1,000 YouTube videos for a human biomechanics dataset. Trained a PyTorch video classifier to detect falls in unconstrained environments working with Prof. Xi Peng.

ENGINEERING EXPERIENCE

Full-Stack Software Engineer Sep. 2021 – May 2024
University of Delaware, PAROT Group *Newark, DE*

- Designed and built two releases of the NSF-funded Atom research portal, providing the scientific community with accessible atomic data.
- Developed the portal using React, Node.js, Next.js, and PostgreSQL. First-author journal publication in *Computer Physics Communications*, 2025.

TECHNICAL SKILLS

Languages	Python, C++, C, Java, JavaScript, TypeScript, HTML, CSS
ML / Vision	PyTorch, OpenCV, NumPy, Pandas, scikit-learn
Web / Databases	React, Node.js, Next.js, PostgreSQL, MongoDB

SELECTED HONORS & AWARDS

- **MIT Summer Research Program Scholar** (Jun. 2023 – Sep. 2023)
- **Google CS Research Mentorship Program Scholar** (Jan. 2022 – Apr. 2022)
- **Eugene du Pont Memorial Scholarship** (2020–2024) — Full-ride scholarship awarded to 1 of 30,000 applicants based on high academic achievement, service, and leadership.
- **Outstanding Senior Student Award** (2024) — Awarded to 1 of 600 senior undergraduates in the CIS department.
- **Rees Carl J & Eleanor K Mathematics Scholarship** (2022) — Awarded to 10 of 200 undergraduates based on high academic achievement in Mathematical Sciences.
- **ICPC Mid-Atlantic Regional Bronze Medal** (2021)

LEADERSHIP & SERVICE

Founder, Competitive Programming Club Feb. 2022 – Sep. 2024
University of Delaware *Newark, DE*

- Founded and grew the club to 100+ members. Organized weekly presentations and programming competitions.

Teaching Assistant, AP Computer Science A Sep. 2021 – May 2023
Microsoft Philanthropies TEALS, Trenton Central High School *Trenton, NJ*

- Supported a class of 16 students through Java presentations and one-on-one help sessions.